bom.xlsx - sheet 'BOM'

Component	Mass (g/cell)	kg/kWh	Formula / split	Source
Positive active material NMC811	7.2826	0.4638	pos coat x 94% (lit. default)	r5_v12_energy.json:layer_kg_m2 x area_m2
Positive conductive additive (carbon black)	0.2324	0.0148	pos coat x 3% (lit. default)	r5_v12_energy.json:layer_kg_m2 x area_m2
Positive binder (PVDF)	0.2324	0.0148	pos coat x 3% (lit. default)	r5_v12_energy.json:layer_kg_m2 x area_m2
Negative active material (graphite)	6.2426	0.3975	neg coat x 94% (lit. default)	r5_v12_energy.json:layer_kg_m2 x area_m2
Negative conductive additive (carbon black)	0.065	0.0041	neg coat x 1% (lit. default)	r5_v12_energy.json:layer_kg_m2 x area_m2
Negative binder (CMC+SBR)	0.1951	0.0124	neg coat x 3% (lit. default)	r5_v12_energy.json:layer_kg_m2 x area_m2
Separator	0.2446	0.0156	L_sep x eps_sep x rho_sep x area	r5_v12_energy.json:layer_kg_m2 x area_m2
Electrolyte	8.9321	0.5688	pore volume x 1.2 g/cm3 (lit. value)	params+geometry pore volume x literature density 1.2 g/cm3
Positive current collector (Al)	2.2183	0.1413	th x 2700 kg/m3 x area	r5_v12_energy.json:layer_kg_m2 x area_m2
Negative current collector (Cu)	5.5212	0.3516	th x 8960 kg/m3 x area	r5_v12_energy.json:layer_kg_m2 x area_m2
Enclosure / case	Not modeled		no shell parameter in Chen2020 set	base params
Tabs / terminals	Not modeled		no tab parameter in Chen2020 set	base params
Total material mass	31.1663	1.9847	stack mass + electrolyte fill	mechanical sum
Cell energy	15.7035		1C discharge energy (V*I time integral)	r5_v12_energy.json:energy_wh